Clustering Method Exploration for Transition State Geometries

Isopropanol on DFTB0 — 74 TS, 131 Methods Tested

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Abstract

We systematically compare 131 clustering configurations across 7 algorithm families (hierarchical, DBSCAN, OPTICS, KMeans, spectral, agglomerative) applied to 74 transition state geometries of isopropanol on DFTB0. The distance metric is aligned RMSD (Kabsch + Hungarian). We evaluate against energy-based ground truth (16 groups), internal quality (silhouette), and cross-method agreement. The central finding is that **RMSD-based structural clustering does not correlate with energy-based grouping** (all ARI < 0.05), meaning structurally similar TS can have very different energies. DBSCAN robustly identifies 7 tight structural clusters, but these are not energy-coherent.

1 Setup

Data: 74 converged TS geometries from NR→GAD search on isopropanol (C_3H_8O, 12 atoms) at 0.2–0.5 Å noise on the DFTB0 PES. Each TS satisfies $n_{\text{neg}} = 1$ and $\|F\| < 0.01$ eV/Å.

Distance metric: Aligned RMSD with Kabsch rotation, Hungarian atom matching (6 equivalent methyl H), and methyl carbon permutation enumeration. All $74 \times 74 = 2701$ unique pairwise distances precomputed.

Ground truth: Energy-based clusters with 0.1 eV tolerance produce 16 groups. The 5 main families (A–E) contain 13–17 members each; the remaining 11 groups are singletons or pairs at unique energies.

Methods tested: 131 configurations:

- Hierarchical (complete, average, single linkage): thresholds 0.1–0.5 Å, $k = 3$–20
- Hierarchical (Ward): on MDS embedding (Ward requires Euclidean input)
- DBSCAN: $\varepsilon = 0.05$–0.5, min_samples $= 2, 3, 5$
- OPTICS: min_samples $= 2, 3, 5$
- KMeans: $k = 3$–20 on MDS embedding
- Spectral: $\sigma =$ median and 25th percentile of RMSD, $k = 3$–10
- Agglomerative (sklearn): complete/average/single, $k = 3$–10

Evaluation metrics:

- Adjusted Rand Index (ARI) vs. energy clusters

- Silhouette score on RMSD matrix
- Calinski–Harabasz on MDS embedding
- Mean within-cluster energy std (lower = more energy-coherent)
- Pairwise ARI between methods (cross-method agreement)

2 Structure vs. Energy: No Correlation

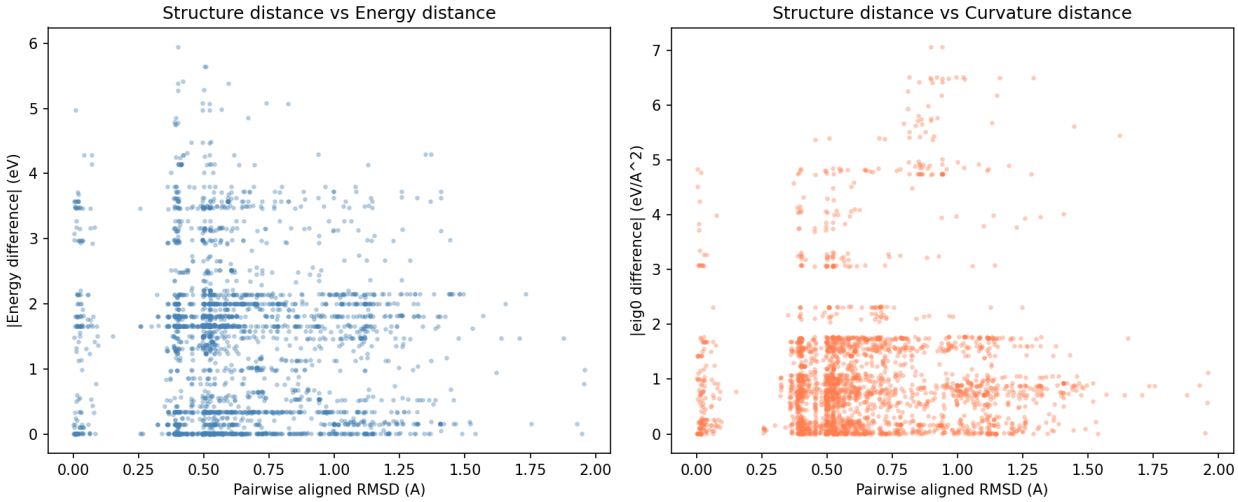

Figure 1: Pairwise aligned RMSD vs. energy difference (left) and curvature difference (right) for all $\binom{74}{2} = 2701$ TS pairs. There is no correlation: pairs at RMSD ~ 0.5 Å span 0–6 eV in energy. Curvature (lowest eigenvalue) shows similarly no correlation with structural distance. This means RMSD alone cannot distinguish chemically distinct TS families.

Figure 1 is the most important result. It shows that **structural similarity (RMSD) and chemical identity (energy) are essentially independent** for this molecule and PES. Two TS with RMSD = 0.3 Å (seemingly similar) can differ by 4 eV, while two TS with RMSD = 1.0 Å (seemingly different) can have the same energy.

Figure 2 confirms this quantitatively: intra-group and inter-group RMSD distributions overlap heavily. The earlier analysis with 67 TS showed a "natural gap" at 0.14–0.26 Å; with 74 TS this gap has filled in, suggesting it was a small-sample artifact.

3 Is There a Natural Cluster Count?

The answer is **no clear natural** k. The silhouette curve (Fig. 3, right) decreases monotonically from $k = 2$ to $k \approx 15$, then rises again as clusters become singletons. DBSCAN consistently finds $k = 7$ (for $\varepsilon = 0.05$–0.25), suggesting 7 tight structural clusters exist, but these account for only 44–48 of 74 TS; the rest are "noise" (26–30 points).

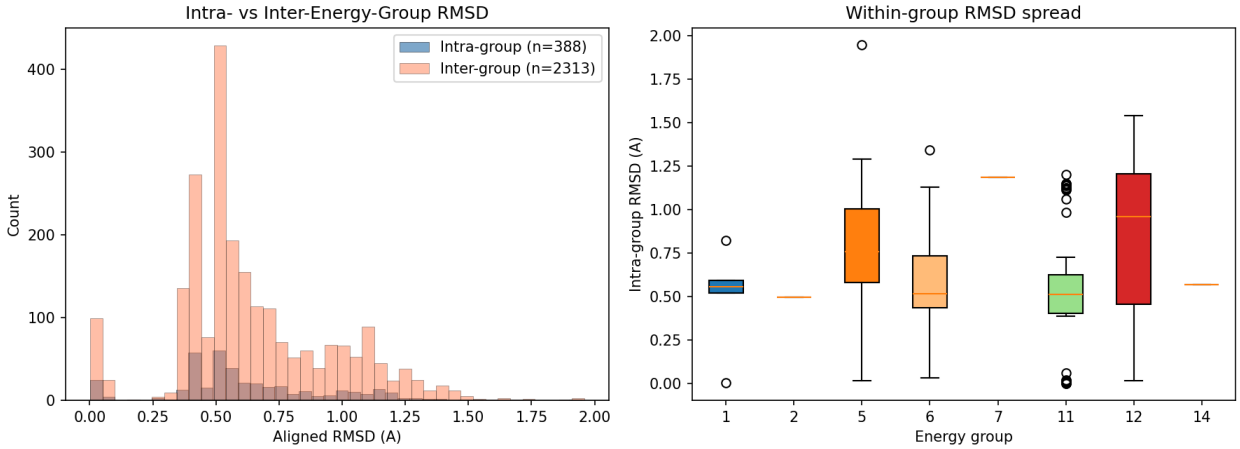

Figure 2: Left: distribution of intra-energy-group and inter-energy-group RMSD. Heavy overlap: the distributions are not separable, so no RMSD threshold cleanly distinguishes same-energy from different-energy TS pairs. Right: within-group RMSD spread per energy group. Most groups have 0.3–1.0 Å internal spread.

4 Cross-Method Agreement

Methods **within** the same family agree reasonably well: hierarchical variants agree at 0.3–1.0 and density-based methods (DBSCAN, OPTICS) agree at ~ 0.6. Methods **across** families show poor agreement (ARI < 0.2). This means the cluster assignments are method-dependent, not an intrinsic property of the data.

5 Dendrogram with Chemical Features

Figure 5 visually demonstrates the RMSD–energy disconnect. Tight structural clusters (low merge height) contain TS with different energies (different colors in the energy bar). This is most visible in the central region (index $\sim$ 30–50) where structurally similar TS span multiple energy families.

6 Embedding Visualizations

7 Quantitative Comparison

Table 1: Best methods by each criterion.

Criterion	Best method	k	Score	Energy std (eV)
ARI vs. energy	spectral (σ=0.56, k=8)	8	0.050	1.08
Silhouette	DBSCAN (ε=0.05, ms= 2)	7	0.940	1.28
Energy coherence	hier. complete (k=8)	8	—	0.70

Even the best energy-coherent method (hierarchical complete, $k = 8$) has within-cluster energy standard deviation of 0.70 eV, which is large given the 6 eV energy range.

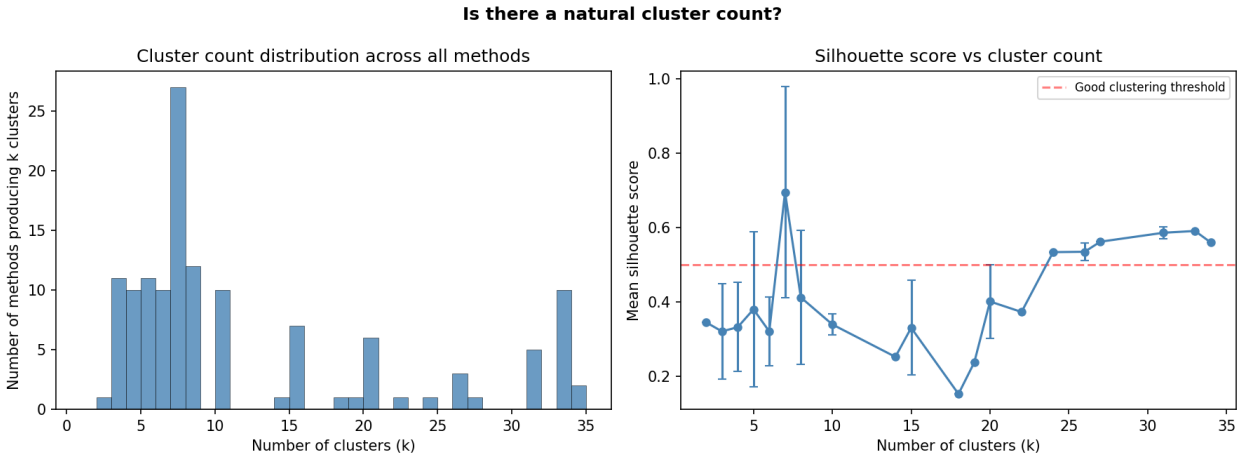

Figure 3: Left: histogram of cluster counts across all 131 methods. Most methods produce $k = 7$ or $k = 31$; no single k dominates. Right: silhouette score vs. k for hierarchical average linkage. Silhouette is highest at $k = 2$ (trivial split) and at high k (~ 29, many singletons). No clear elbow suggests there is no single "correct" number of structural clusters.

8 Conclusions

1. **RMSD clustering $\neq$ energy clustering.** All 131 RMSD-based methods achieve ARI < 0.05 against energy-based ground truth. Structural similarity and chemical identity are decoupled for isopropanol TS on DFTB0.

2. **No natural structural cluster count.** Silhouette analysis shows no elbow; DBSCAN robustly finds 7 tight clusters but classifies 30/74 TS as noise.

3. **Methods agree within families, not across.** Hierarchical methods agree with each other (ARI $\sim$ 0.3–1.0), but not with DBSCAN or spectral (ARI < 0.2).

4. **For diffusion model training**, TS should be classified by energy (or energy + eigenvalue), not by RMSD. The 5 main energy families (A–E) are the chemically meaningful grouping.

5. **RMSD clustering is still useful** for identifying structural sub-variants (H-rotamers, mirror images) *within* an energy family, but not for distinguishing families.

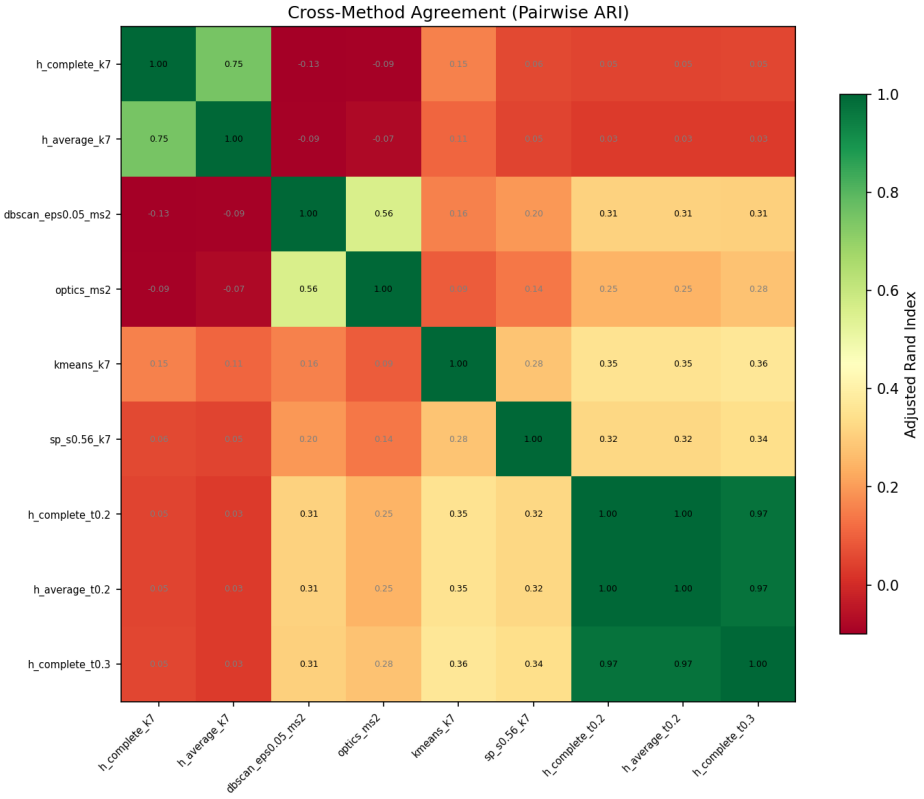

Figure 4: Pairwise ARI between representative methods from each algorithm family. Hierarchical complete and average linkage at threshold 0.2 Å agree perfectly (ARI = 1.0). Hierarchical methods agree moderately with each other (0.3–0.5). DBSCAN/OPTICS form their own block (ARI $\sim$ 0.6). Spectral and KMeans show low agreement with everything (< 0.2). Overall mean ARI = 0.25.

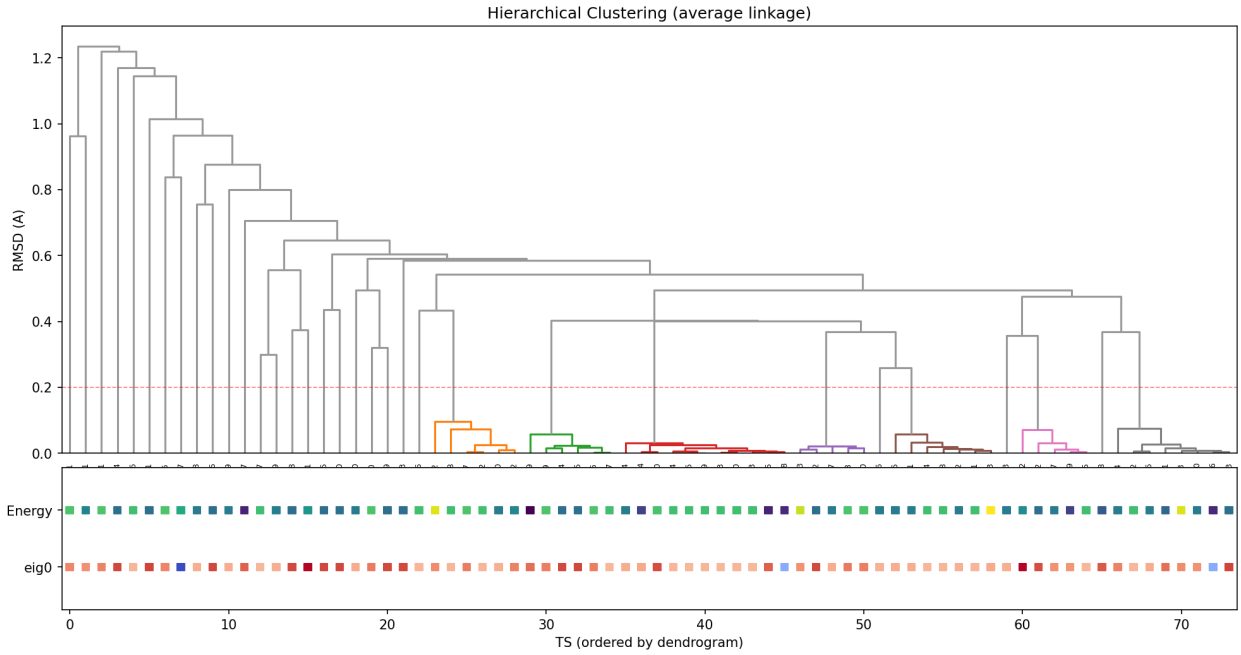

Figure 5: Hierarchical clustering dendrogram (average linkage) with energy (top color bar) and lowest eigenvalue (bottom color bar) mapped onto the leaf order. Adjacent leaves in the dendrogram often have different energy colors, confirming that structural proximity $\neq$ energy proximity. The red dashed line at 0.2 Å is the original clustering threshold.

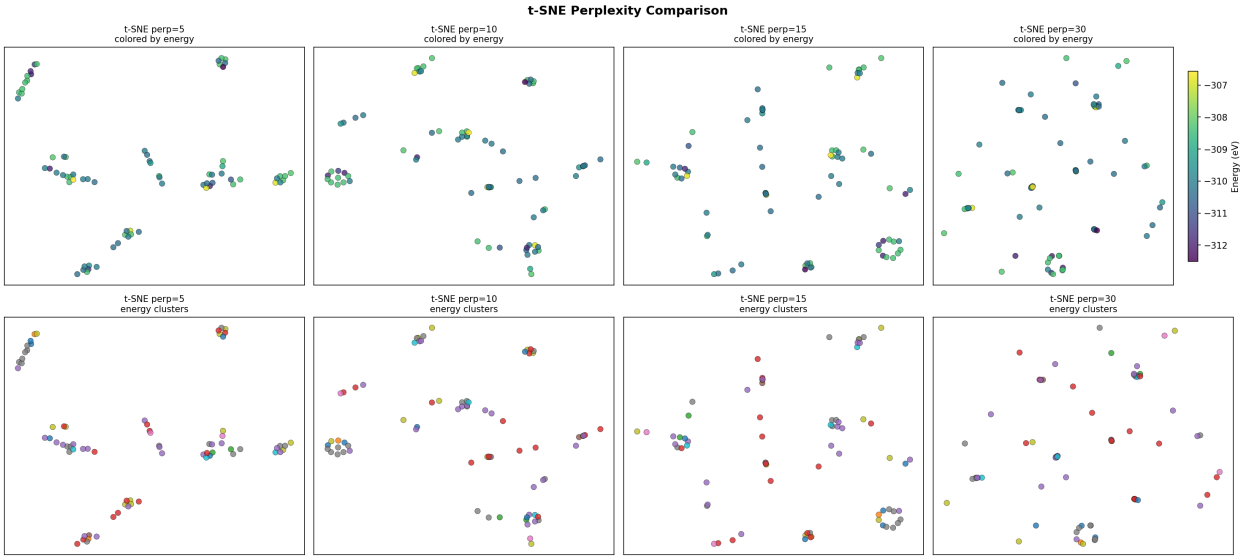

Figure 6: t-SNE embeddings at perplexity 5, 10, 15, 30 (columns). Top: colored by energy. Bottom: colored by energy cluster assignment. Low perplexity (5) fragments into tight micro-clusters that mix energies. High perplexity (30) shows broader structure but still no clean energy separation. Perplexity 15 provides the best balance.

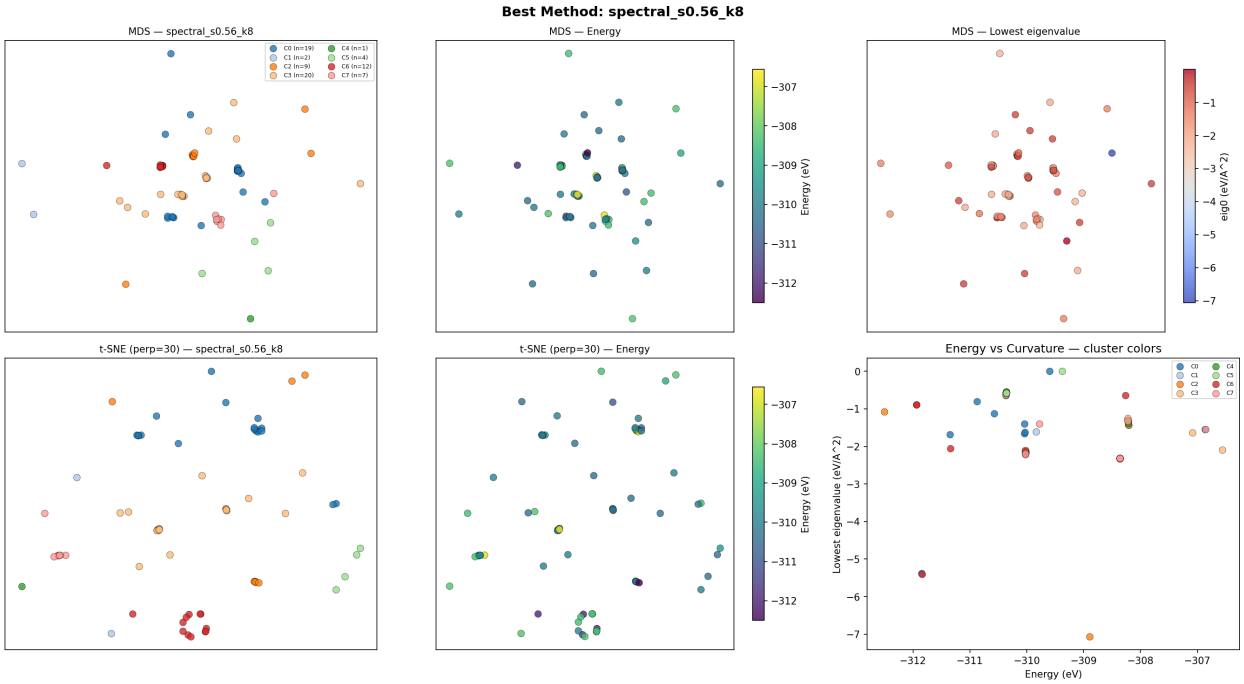

Figure 7: The "best" RMSD-based method (spectral, $\sigma = 0.56$, $k = 8$; ARI = 0.05). Top row: MDS embedding colored by cluster assignment, energy, and lowest eigenvalue. Bottom row: t-SNE (perp= 30) colored by cluster and energy, plus energy vs. curvature scatter. Even the best method produces clusters that span 2+ eV in energy.

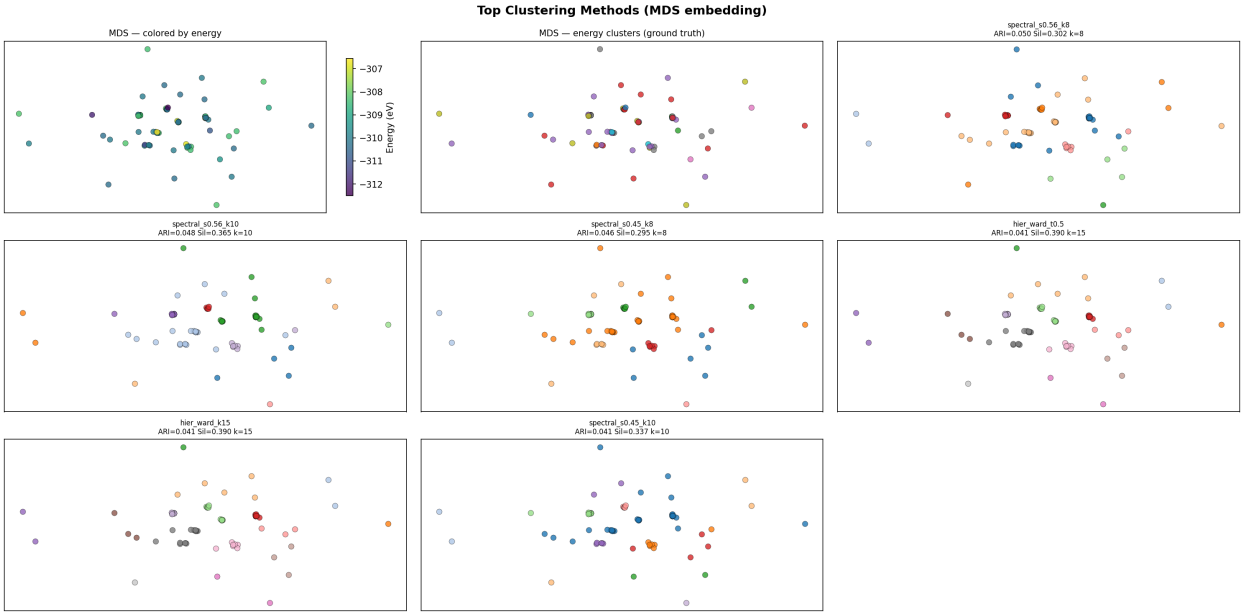

Figure 8: MDS embedding colored by the top 6 clustering methods (ranked by ARI). First two panels show energy coloring and energy cluster ground truth. All methods produce visually different cluster assignments, and none matches the energy coloring.

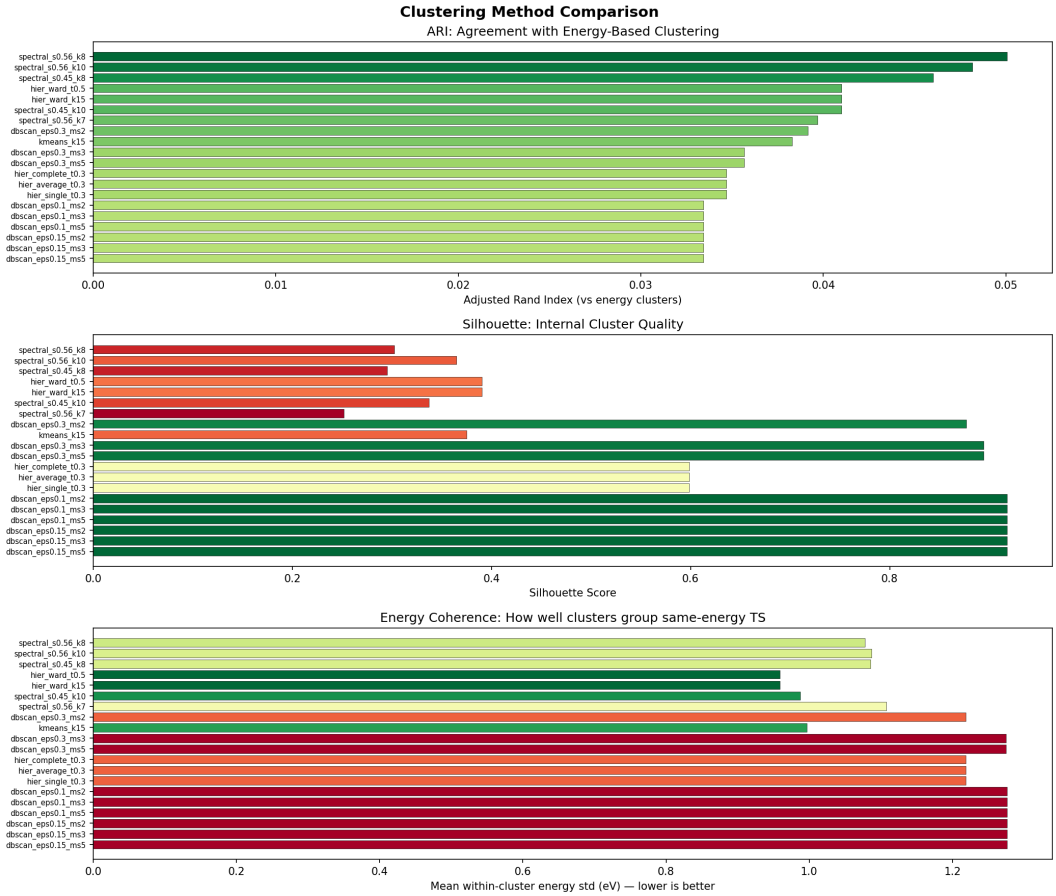

Figure 9: Top 20 methods ranked by ARI (top), silhouette (middle), and energy coherence (bottom). Methods that score well on silhouette (DBSCAN, high k) score poorly on energy coherence, and vice versa. No method scores well on all three metrics simultaneously.